

LESSON 3

MEASURES OF RELATIVE POSITION

Overview:

This lesson will focus on locating the relative position of a data value in a given data set. The measures of relative position are in addition to measures of central tendency and measures of dispersion. These measures include standard scores, quartiles, and percentiles.

Learning Outcomes:

- At the end of Lesson 3, the students can:
- 1. compare the z-scores of two data sets;
 - 2. apply z-scores in solving related to real life problems;
 - 3. identify the pth percentile and quartiles of the different data values; and

Materials Needed:

- statistical data worksheets
- slide presentation
- Calculator

Duration: 1.5 hours

Learning Contents:

3.1 Z – SCORE

The **z-score** or the standard score for a given data value x is the number of standard deviations that x is above or below the mean of the data. The formula is

$$z = \frac{\text{value} - \text{mean}}{\text{standard deviation}}$$

z-score of a sample

$$z = \frac{X - \bar{X}}{s}$$

z-score of a population

$$z = \frac{X - \mu}{\sigma}$$

Examples

1. Find the z score for each test, and state which is higher.

Test A	$X = 38$	$\bar{X} = 40$	$s = 5$
Test B	$X = 94$	$\bar{X} = 100$	$s = 10$

Solution

For Test A,

$$z = \frac{X - \bar{X}}{s} = \frac{38 - 40}{5} = -0.4$$

For Test B,

$$z = \frac{X - \bar{X}}{s} = \frac{94 - 100}{10} = -0.6$$

The score for test A is relatively higher than the score for test B.

2. A consumer group tested a sample of 100 light bulbs. It was found that the mean life expectancy of the bulbs was 842 hours, with a standard deviation of 90. One particular light bulb from the DuraBright Company had a z-score of 1.2. what was the life span of this light bulb?

Solution

Substitute the given values into the z-score equation and solve for X .

$$z = \frac{X - \bar{X}}{s}, z = 1.2, \bar{X} = 842, S = 90$$

$$1.2 = \frac{X - 842}{90}$$

$$108 = X - 842$$

$$X = 108 + 842$$

$$X = 950$$

Thus, the light bulb had a life span of 950 hours.

3.2 PERCENTILES

Percentiles divide the data set into 100 equal groups. A value x is the p^{th} **percentile** of a data values that are less than x .

Percentile Formula

Given a set of data and a data value x ,

$$\text{Percentile of score } x = \frac{\text{number of data values less than } x + 0.5}{\text{total number of data values}} \cdot 100\%$$

Example

3. A teacher gives a 20-point test to 10 students. The scores are shown here.
Find the percentile rank of a score of 12.

18, 15, 12, 6, 8, 2, 3, 5, 20, 10

Solution

Arrange the data in order from lowest to highest.

2, 3, 5, 6, 8, 10, 12, 15, 18, 20

Then substitute into the formula.

$$\text{Percentile} = \frac{\text{number of data values less than } x + 0.5}{\text{total number of data values}} \cdot 100$$

Since there are six values below a score of 12, the solution is

$$\text{Percentile} = \frac{6+0.5}{10} \cdot 100\% = 65^{\text{th}} \text{ percentile}$$

Thus, a student whose score was 12 did better 65% of the class.

Example

4. In a recent year, the median annual population of a region was 74,840. If the 90th percentile for the annual population of a region was 105,900, find the percent of regions whose annual population was
 - a. more than 74,840;
 - b. less than 105,900;
 - c. between 74,840 and 105,900.

Solution

- a. By definition, the median is the 50th percentile. Therefore, 50% of the region’s population was more than 74,840.
- b. Because 105,900 is the 90th percentile, 90% of the region’s population was less than 105,900.
- c. From parts a and b, 90% – 50% = 40% of the regions’ population was between 74,480 and 105,900.

3.3 QUARTILES

Quartiles divide the distribution into four groups, separated by Q_1 , Q_2 , and Q_3 . Note that Q_1 is the same as the 25th percentile; Q_2 is the same as the 50th percentile, or the median; Q_3 corresponds to the 75th percentile, as shown:

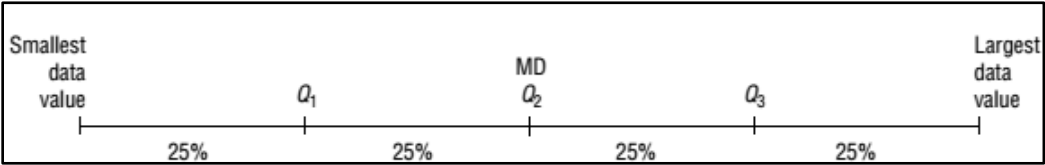


Figure 3.1

Procedure Table	
Finding Data Values Corresponding to Q_1 , Q_2 , and Q_3 .	
Step 1	Arrange the data in order from lowest to highest.
Step 2	Find the median of the data values. This is the value for Q_2 .
Step 3	Find the median of the data values that fall below Q_2 . This is the value for Q_1 .
Step 4	Find the median of the data values that fall above Q_2 . This is the value for Q_3 .

Example

5. Find Q_1 , Q_2 , and Q_3 for the data set 15, 13, 6, 5, 12, 50, 22, 18.

Solution

Step 1 Arrange the data in order from lowest to highest.

5, 6, 12, 13, 15, 18, 22, 50

Step 2 Find the median (Q_2).

5, 6, 12, 13, 15, 18, 22, 50
 ↑
 Median

$$MD = \frac{13+15}{2} = 14$$

Step 3 Find the median of the data values less than 14.

$$5, 6, \underset{\substack{\uparrow \\ Q_1}}{12}, 13$$

$$Q_1 = \frac{6+12}{2} = 9$$

Step 4 Find the median of the data values greater than 14.

15, 18, 22, 50
 $\uparrow$
 Q_3

$$Q_3 = \frac{18+22}{2} = 20$$

Hence, $Q_1 = 9$, $Q_2 = 14$, and $Q_3 = 20$.

Learning Activity:

Consider the scores of ten college students in a test. Complete the table below by computing the **z-score** and **pth percentile** of every score. Find also the **first quartile (Q_1)**, **second quartile (Q_2)**, and **third quartile (Q_3)** of the given data set.

Hint: Solve for the mean and standard deviation for z-score.

Scores	z-score	p th percentile
25		
36		
42		
38		
27		
20		
41		

33		
30		
44		
$Q_1 =$ _____	$Q_2 =$ _____	$Q_3 =$ _____

Learning Evaluation:

- I. Determine the z-scores of the following data sets and round each z-score to the nearest hundredth.
 1. A data set has a mean of $\bar{x} = 75$ and a standard deviation of 11.5. Find the z-score for each of the following.

- a. $x = 85$
 - b. $x = 95$

- c. $x = 50$
 - d. $x = 75$
 2. A data set has a mean of $\bar{x} = 212$ and a standard deviation of 40. Find the z-score for each of the following.

- a. $x = 220$
 - b. $x = 224$

- c. $x = 300$
 - d. $x = 100$
- II. Use the sets of data relating to examination scores of 11 candidates in Mathematics and English to answer the following:
 1. What is the 10th percentile of the Mathematics scores?
 2. Find the 25th percentile of the English scores.
 3. Determine the 75th percentile of the Mathematics scores.

Student	A	B	C	D	E	F	G	H	I	J	K
Mathematics	62	91	43	31	57	63	80	37	43	5	78
English	65	57	55	37	62	70	73	49	65	41	64

References:

Aufman, R. et. al. (2018). *Mathematics in the Modern World*. Rex Book Store Inc., Sampaloc, Manila, Philippines.

Brase,C. et.al. (2018). *Understanding Basic Statistics*. 8th Edition. Cengage Learning.

Nalangan, C. et. al. (2009). *Laboratory Manual in Statistics 1 (Elementary Statistics)*. Rex Book Store Inc.